The worker is the constant-branch slice of the weld

A bicomodule comparison of state-entanglement and data-aggregation

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Abstract

In the polynomial-functor semantics of stateful agents, two constructions package the state of a worker: the *weld* $B = \mathsf{St}S \triangleleft \mathsf{St}T$, which composes two costate comonads, and the *worker grading* $A = \mathsf{St}(S \times T)$, which tensors them. It was recently established that B is a genuine $(\mathsf{St}T, \mathsf{St}S)$-bicomodule whose represented parametric-right-adjoint functor is $\bar{F}_B(V) = (T \times V)^S$. The status of the worker A inside this bicomodule was left open, and a natural conjecture — that A is the right-adjoint "content" factor of B — had resisted a clean verdict, because A and B live at different levels of the framed bicategory $\mathsf{Cat}^\sharp$ and so cannot be compared as written. We resolve this by reading A in the same variance as B: the worker is canonically the *constant-branch sub-bicomodule* $B' \subseteq B$, with carrier polynomial exactly A. Worker and weld then *share* the content power $(-)^S$ and differ *only* in their coproduct-aggregation index — the constant branch maps T for the worker against all branch maps T^S for the weld. The separation is proper precisely on $|S|, |T| \geq 2$, and degenerates (worker = weld on the nose) when $|S| = 1$ or $|T| = 1$. State-entanglement and data-aggregation are thus genuinely different collapses of one bicomodule.

1 Introduction

The problem. A stateful agent — a *worker* — is modelled in the container/polynomial program as an interface together with a store of internal state. When a worker with input set S and output set T carries its state, two polynomial constructions compete to express "state over the interface". The first is the *weld*

$$B \;=\; \mathsf{St}S \triangleleft \mathsf{St}T,$$

the composition-product ($\triangleleft$) of the costate comonads of S and T; this is the Zappa–Szép style orchestration weld, in which S runs and then T runs. The second is the *worker grading*

$$A \;=\; \mathsf{St}S \otimes \mathsf{St}T \;=\; \mathsf{St}(S \times T),$$

the Dirichlet tensor of the two comonads, which fuses the two states into a single store on $S \times T$. Both are comonoids — small categories — in $(\mathsf{Poly}, \triangleleft, \mathsf{y})$, and both claim to be "the state of the composite worker". Which is right, and how are they related?

The gap. The question acquired teeth once the weld was placed inside a framed bicategory. By the Spivak–Garner–Fairbanks correspondence [1], the framed bicategory $\mathsf{Cat}^\sharp$ of categories and parametric right adjoints is equivalent to $\mathsf{Comod}(\mathsf{Poly})$, so a horizontal 1-cell $c \to d$ is a *bicomodule*, equivalently a parametric right adjoint (PRA) functor $c\text{-}\mathsf{Set} \to d\text{-}\mathsf{Set}$. A companion note established three facts about the weld, of which the first was independently refereed and confirmed:

(1) *(existence, peer-reviewed)* $B = \mathsf{St}S \triangleleft \mathsf{St}T$ carries a genuine $(\mathsf{St}T, \mathsf{St}S)$-bicomodule structure, hence is a PRA functor $\mathsf{St}T\text{-}\mathsf{Set} \to \mathsf{St}S\text{-}\mathsf{Set}$, with

$$\bar{F}_B(V) \;\cong\; (T \times V)^S \;=\; T^S \times V^S;$$

(2) *(identification)* the worker A is the right-adjoint "content" factor of B under the SGF slice factorization;

(3) *(tail)* the data discarded by the worker's retract is the coproduct aggregation tail of B.

Clauses (2) and (3) were the live question, and the obstacle was a *type error*: the worker A is a priori a 0-cell of $\mathsf{Cat}^\sharp$ (a comonoid, a category), while the weld content is a 1-cell (a bicomodule $\mathsf{St}T \to \mathsf{St}S$). One cannot ask whether a category "equals" the content of a bicomodule until one has read the category as a bicomodule of the same variance. Until that reading is fixed, any comparison of exponents is comparing objects at different levels — and indeed the naive comparison gives the wrong locus, declaring worker $\neq$ content already at $|S| = 1$, where in fact $B = A$ on the nose.

The result. We supply the missing reading and settle (2) and (3) together. The worker has a canonical $(\mathsf{St}T, \mathsf{St}S)$-bicomodule structure *inside* the weld: it is the sub-bicomodule carried by the *constant* branch maps.

Theorem 1 (Worker = constant-branch slice; see Proposition 1 and Theorems 5, 6)**.** *Let S, T be finite nonempty sets. Read the worker $A = \mathsf{St}(S \times T)$ as a $(\mathsf{St}T, \mathsf{St}S)$-bicomodule via the inclusion $\sigma\colon A \xrightarrow{\sim} B'$ onto the constant-branch sub-bicomodule $B' \subseteq B$ ($\sigma_1(s,t) = (s, \mathrm{const}_t)$, $\sigma^\sharp = \mathrm{id}_{S\times T}$). Then*

$$\bar{F}_A(V) \;=\; \textstyle\sum_{g \in \mathrm{const}(T)} V^S \;=\; T \cdot V^S \qquad \subseteq \qquad \bar{F}_B(V) \;=\; \textstyle\sum_{g \in T^S} V^S \;=\; T^S \cdot V^S.$$

Worker and weld share *the right-adjoint content power $(-)^S$; they differ* only *in the coproduct-aggregation index, $T \cong \mathrm{const}(T)$ for the worker against all branch maps $T^S = \mathrm{Hom}(S,T)$ for the weld. The inclusion $\bar{F}_A \hookrightarrow \bar{F}_B$ is an equality iff $|S| = 1$ or $|T| = 1$, and is proper precisely on $|S| \geq 2 \wedge |T| \geq 2$.*

Thus clause (2) is **false** where it is interesting ($|S|, |T| \geq 2$) — but *not* for the reason first proposed. The content *power* $(-)^S$ is in fact *shared*; what the worker fails to be is the whole prafunctor, because it aggregates over the constant branch maps alone. And clause (3), once stated honestly, splits into a true half (the aggregation tail *is* the full branch-map set T^S) and a false half (the retract does *not* discard "the non-constant maps").

The method. The one idea is Definition 3: inside the weld's position set $S \times T^S$, the positions with *constant* branch map form a sub-polynomial whose carrier is exactly A, and the weld's two coactions preserve constancy because they never touch the branch-map coordinate. Everything else is bookkeeping on the peer-reviewed formula $\bar{F}_B(V) = (T \times V)^S$: restrict its coproduct index from T^S to $\mathrm{const}(T)$ (Section 5), and read off what the worker's retract keeps and discards (Section 6).

Consequences. The two constructions carry a genuine semantic distinction that the orchestration calculus must respect. State-*entanglement* (the worker A) records one branch choice per output token $t \in T$; data-*aggregation* (the weld B) records one branch choice per response policy $g \in T^S$. Both read the whole input state — the shared content $(-)^S$ — but the weld aggregates over all policies and the worker only over the constant ones. The exact locus where the distinction bites is $|S|, |T| \geq 2$: with a single input or a single output there is nothing to entangle, and $A = B$.

Status and outline. This is a referee-response revision: clause (1) and the formula $\bar{F}_B(V) = (T \times V)^S$ are peer-reviewed input, used but not re-litigated. Section 2 fixes conventions (bicomodules, their handedness, the PRA correspondence, the costate comonad). Section 3 recalls the clause-(1) core. Section 4 proves the new content: the worker is the constant-branch subbicomodule. Section 5 carries out the corrected comparison, Section 6 the corrected retract analysis. Section 7 states the unified result and its orchestration reading. Section 8 is a point-by-point response to the referee.

2 Preliminaries

We work in the category Poly of polynomial functors $\mathsf{Set} \to \mathsf{Set}$ and natural transformations, embedded fully faithfully in $[\mathsf{Set}, \mathsf{Set}]$ [1, 2]. A polynomial $p = \sum_{i \in p(1)} \mathsf{y}^{p[i]}$ has a set of *positions* $p(1)$ and, at each position i, a *direction set* $p[i]$; it acts by $p(X) = \sum_i X^{p[i]}$. The *composition product* is functor composition $(p \triangleleft q)(X) = p(q(X))$ with unit y; the *Dirichlet product* $\otimes$ has positions $p(1) \times q(1)$, directions $p[i] \times q[j]$, and unit y. A morphism $\phi\colon p \to q$ is a function $\phi_1\colon p(1) \to q(1)$ on positions together with a backward function $\phi^\sharp\colon q[\phi_1 i] \to p[i]$ on directions, for each i.

Comonoids are categories. A comonoid in $(\mathsf{Poly}, \triangleleft, \mathsf{y})$ is exactly a small category [1, Prop. 5.2]: the positions are objects, the directions at an object are the morphisms out of it, the counit selects identities and the comultiplication is composition. We write $\mathsf{Cat}^\sharp$ for the framed bicategory whose objects are these polynomial comonoids.

Definition 1 (Bicomodule; [1, Def. 4.2, Rmk 4.3])**.** Let $(c, \varepsilon_c, \delta_c)$ and $(d, \varepsilon_d, \delta_d)$ be comonoids in $(\mathsf{Poly}, \triangleleft, \mathsf{y})$. A (c, d)-*bicomodule* $m\colon c \to d$ is a carrier $m \in \mathsf{Poly}$ with a left d-coaction and a right c-coaction

$$d \triangleleft m \xleftarrow{\lambda} m \xrightarrow{\rho} m \triangleleft c$$

satisfying counit, coassociativity, and compatibility laws. (The variance follows the Leibniz reading of [1, Rmk 4.3]: λ sends m to "d after m" and ρ to "m after c", so m goes from c to d.)

Theorem 2 (SGF correspondence; [1, Prop. 3.9, Thm. 4.28])**.** *For comonoids c, d, the framed bicategories $\mathsf{Cat}^\sharp$ and $\mathsf{Comod}(\mathsf{Poly})$ are equivalent; in particular*

$$\mathsf{Comod}(\mathsf{Poly})(c, d) \;\simeq\; \mathsf{PRA}\big(c\text{-}\mathsf{Set},\; d\text{-}\mathsf{Set}\big),$$

so a (c, d)-bicomodule is *a parametric right adjoint functor c-$\mathsf{Set} \to d$-Set. For a c-variable polynomial F, the SGF slice factorization writes the PRA as a coproduct of representables indexed by $F(1)$, with right-adjoint content given by the connected-limit part [1, Prop. 3.6].*

Definition 2 (Costate comonad; [1, Ex. 5.6])**.** For a nonempty set S write

$$\mathsf{St}S \;=\; S \cdot \mathsf{y}^S \;=\; \left[\begin{smallmatrix} S \\ S \end{smallmatrix}\right],$$

the *costate* (or store) comonad, i.e. the comonad induced by the adjunction $S \times (-) \dashv (-)^S\colon \mathsf{Set} \to \mathsf{Set}$. It has S positions, each with direction set S. As a category it is the *indiscrete category* $\mathbb{I}(S)$ on object set S: a unique morphism between any two objects.

Throughout, S, T are finite and nonempty. We reserve the symbol Δ for the diagonal functor and never for a comonoid; the two costate comonoids are $\mathsf{St}S$ and $\mathsf{St}T$, and the worker carrier is $\mathsf{St}(S \times T)$.

Lemma 1 (The indiscrete collapse)**.** *For $S \neq \emptyset$ the indiscrete category $\mathbb{I}(S)$ is equivalent to the terminal category $\mathbf{1}$; hence $\mathsf{St}S$-$\mathsf{Set} \simeq \mathsf{Set}$, and under this equivalence every copresheaf X has $\lim X \cong X(s_0) \cong \operatorname{colim} X$ for each object s_0, and the representable at any object is the terminal copresheaf.*

Proof. In $\mathbb{I}(S)$ there is a unique isomorphism between any two objects, so the inclusion $\mathbf{1} \to \mathbb{I}(S)$ of any object is an equivalence. Equivalent categories have equivalent copresheaf categories, and limits/colimits transport accordingly. □

We use $\bar{F}$ for the endofunctor of Set obtained by transporting a store-bicomodule's PRA functor across the collapses $\mathsf{St}S\text{-}\mathsf{Set} \simeq \mathsf{Set}$, $\mathsf{St}T\text{-}\mathsf{Set} \simeq \mathsf{Set}$; right-adjointness and the SGF factorization are equivalence-invariant.

3 The weld bicomodule (peer-reviewed core)

We recall the clause-(1) results, which are input to what follows. The naming departs from the companion note in one respect: $B = \mathsf{St}S \triangleleft \mathsf{St}T$ is *not* a regular bicomodule (the regular bicomodule of a comonoid c is c as a (c, c)-bicomodule). It is the *unit-composite* bicomodule — the composite over the unit y of the two one-sided regular comodules — as Theorem 4 makes explicit.

Lemma 2 (Unit-composite bicomodule)**.** *For any comonoids c, d in a monoidal category $(\mathcal{V}, \triangleleft, \mathsf{y})$, the object $d \triangleleft c$ is a (c, d)-bicomodule with*

$$\lambda \;=\; \delta_d \triangleleft c \colon d \triangleleft c \to d \triangleleft d \triangleleft c, \qquad \rho \;=\; d \triangleleft \delta_c \colon d \triangleleft c \to d \triangleleft c \triangleleft c.$$

Proof. The left laws (counit, coassociativity) are the comonoid laws of d whiskered on the right by $\triangleleft c$; the right laws are the comonoid laws of c whiskered on the left by $d\triangleleft$. For compatibility, λ acts on the first (d) factor and ρ on the last (c) factor; since $\triangleleft$ is a bifunctor, the two whiskerings commute and both composites equal $\delta_d \triangleleft \delta_c$. □

Theorem 3 (Clause (1), peer-reviewed)**.** *The weld $B = \mathsf{St}S \triangleleft \mathsf{St}T$ is the unit-composite $(\mathsf{St}T, \mathsf{St}S)$-bicomodule (take $c = \mathsf{St}T$, $d = \mathsf{St}S$, carrier $d\triangleleft c = \mathsf{St}S\triangleleft\mathsf{St}T$), with $\lambda = \delta_{\mathsf{St}S}\triangleleft\mathsf{St}T$ and $\rho = \mathsf{St}S \triangleleft \delta_{\mathsf{St}T}$. Hence B is a horizontal 1-cell of $\mathsf{Cat}^\sharp$, i.e. a PRA functor $\mathsf{St}T\text{-}\mathsf{Set} \to \mathsf{St}S\text{-}\mathsf{Set}$.*

Proof. Lemma 2 with $c = \mathsf{St}T$, $d = \mathsf{St}S$; both are comonoids, and Theorem 2 supplies the PRA reading. □

Theorem 4 (The represented prafunctor; peer-reviewed)**.** *After the collapse of Lemma 1,*

$$\bar{F}_B(V) \;\cong\; (T \times V)^S \;=\; T^S \times V^S.$$

As a polynomial endofunctor of Set, $\bar{F}_B = T^S \cdot \mathsf{y}^S$: its position set (the coproduct index $\bar{F}_B(1)$) is the branch-map set $T^S = \mathrm{Hom}(S, T)$, and its direction set is S.

Proof. The unit-composite factors over y as

$$\mathsf{St}T \xrightarrow{G} \mathsf{y} \xrightarrow{H} \mathsf{St}S, \qquad H \triangleleft G \;=\; \mathsf{St}S \triangleleft \mathsf{St}T \;=\; B,$$

where G is the right $\mathsf{St}T$-comodule with carrier $\mathsf{St}T$ and $\rho = \delta_{\mathsf{St}T}$, and H is the left $\mathsf{St}S$-comodule with carrier $\mathsf{St}S$ and $\lambda = \delta_{\mathsf{St}S}$. Bicomodule composition over y is just $\triangleleft$, so $F_B = F_H \circ F_G$.

For $F_G \colon \mathsf{St}T\text{-}\mathsf{Set} \to \mathsf{Set}$, the right $\mathsf{St}T$-comodule with carrier $\mathsf{St}T$ is

$$F_G(X) \;=\; \sum_{t \in T} \mathsf{St}T\text{-}\mathsf{Set}(\mathsf{y}_t, X),$$

where y_t denotes the representable at the object t. By Lemma 1 each y_t is terminal, so the hom-set is $\lim X$; hence $F_G(X) = \sum_{t \in T} \lim X$ and, after transport, $\bar{G}(V) = T \times V$.

For $F_H \colon \mathsf{Set} \to \mathsf{St}S\text{-}\mathsf{Set}$, a left $\mathsf{St}S$-comodule is a diagram $\mathsf{St}S \to \mathsf{Poly}$ [1, Prop. 5.7]; the regular one assigns to each object $a \in S$ the component polynomial y^S (one position, S directions, read off $\delta_{\mathsf{St}S}$). So $F_H(W)(a) = (\mathsf{y}^S)(W) = W^S$, and after transport $\bar{H}(W) = W^S$.

Therefore $\bar{F}_B(V) = \bar{H}(\bar{G}(V)) = (T \times V)^S$, and $(T \times V)^S = \{S \to T \times V\} = \{(g, h) : g \colon S \to T,\ h \colon S \to V\} = T^S \times V^S$. □

By Theorem 2 the SGF slice factorization of $\bar{F}_B(V) = \sum_{g \in T^S} V^S$ has coproduct ("Σ") index $\bar{F}_B(1) = T^S$ and right-adjoint *content* the power $(-)^S$ (represented by the direction set S). This content split is the backbone of the comparison below.

Remark 1. The represented prafunctor $\bar{F}_B = T^S \mathsf{y}^S$ is neither the underlying polynomial of $A = (S \times T)\mathsf{y}^{S \times T}$ nor that of the weld carrier $B = (S \cdot T^S)\mathsf{y}^{S \times T}$. The PRA functor of a bicomodule is not the underlying functor of its carrier — the trap at the heart of the problem. (At $|S| = |T| = 2$, $V = 2$: $\bar{F}_B = 16$, $A_{\text{und}} = 64$, $B_{\text{und}} = 128$.)

4 The worker as a bicomodule: the constant-branch slice

Here is the new content. To compare the worker A with the weld content we must read A as a $(\mathsf{St}T, \mathsf{St}S)$-bicomodule of the *same* variance as B. There is a canonical such reading, and it sits visibly inside B.

Recall the weld carrier as a polynomial: $(\mathsf{St}S \triangleleft \mathsf{St}T)(X) = S \times (T \times X^T)^S = S \times T^S \times X^{S \times T}$, so B has position set $S \times T^S$ — a current input $s \in S$ and a branch map $g \colon S \to T$ — and direction set $S \times T$ at each position.

Definition 3 (Constant-branch slice)**.** Let $\mathrm{const} \colon T \hookrightarrow T^S$, $t \mapsto \mathrm{const}_t$, be the inclusion of the constant branch maps. Let $B' \subseteq B$ be the sub-polynomial on the positions with constant branch map,

$$B' \;=\; \big\{(s, g) \;:\; g \in \mathrm{const}(T)\big\},$$

with inherited direction set $S \times T$. Its position set is $S \times \mathrm{const}(T) \cong S \times T$, so as a polynomial

$$B' \;=\; (S \times T) \cdot \mathsf{y}^{S \times T} \;=\; \mathsf{St}(S \times T) \;=\; A$$

on the nose.

Lemma 3 (B' is a sub-bicomodule)**.** *B' is closed under the weld coactions $\lambda = \delta_{\mathsf{St}S} \triangleleft \mathsf{St}T$ and $\rho = \mathsf{St}S \triangleleft \delta_{\mathsf{St}T}$; hence $(B', \lambda|, \rho|)$ is a $(\mathsf{St}T, \mathsf{St}S)$-sub-bicomodule of B.*

Proof. Both coactions leave the branch-map coordinate $g = (t_s)_{s \in S}$ *pointwise unchanged*, only duplicating or reorganizing positions.

- $\rho = \mathsf{St}S \triangleleft \delta_{\mathsf{St}T}$ applies $\delta_{\mathsf{St}T}$ inside each of the S slots; since $\delta_T(t, \psi) = (t,\ j \mapsto (j, \psi))$ keeps the first coordinate t, the inner copy of B has the *same* g. A constant g stays constant.
- $\lambda = \delta_{\mathsf{St}S} \triangleleft \mathsf{St}T$ duplicates the whole $\mathsf{St}T$-ground: since $\delta_S(s, \varphi) = (s,\ i \mapsto (i, \varphi))$, every inner copy is (i, φ) with the *same* φ, hence the same g. Constant stays constant.

The counit and (co)associativity laws restrict because they are the comonoid laws of $\mathsf{St}S, \mathsf{St}T$ whiskered, and B' is a full sub-on-positions with inherited directions. □

Proposition 1 (The worker's canonical reading)**.** *The map $\sigma \colon A \to B$ with $\sigma_1(s, t) = (s, \mathrm{const}_t)$ on positions and $\sigma^\sharp = \mathrm{id}_{S \times T}$ on directions is an isomorphism of $(\mathsf{St}T, \mathsf{St}S)$-bicomodules $A \xrightarrow{\sim} B'$.*

Proof. σ_1 is the bijection $S \times T \cong S \times \mathrm{const}(T)$ onto the B'-positions, and $\sigma^\sharp = \mathrm{id}_{S \times T}$ matches direction sets. By Lemma 3 the coactions act on B' exactly as they do on A under this identification (both leave the branch value alone), so σ commutes with λ and ρ. A position-injection with $\sigma^\sharp = \mathrm{id}$ is an isomorphism onto its image B'. □

Remark 2 (honesty). There is a second, conceptual description of this reading: A carries the $(\mathsf{St}T, \mathsf{St}S)$-bicomodule structure obtained by cobase change of the regular (A, A)-bicomodule along the comonoid projections $A \to \mathsf{St}S$ (left) and $A \to \mathsf{St}T$ (right) induced by the set

projections of $S \times T$. The sub-bicomodule description of Definition 3 is primary and is what the comparison uses; the cobase-change description is the justification that B' is the *canonical* reading rather than an arbitrary sub-object, and it agrees with B' on carriers and positions. A fully formal coaction-by-coaction identification of the two structures for general S, T is not written out here, as it is not needed: the comparison uses only $A \cong B'$.

5 The comparison: a coproduct-index separation

Theorem 5 (Corrected comparison)**.** *Read A and B as $(\mathsf{St}T, \mathsf{St}S)$-bicomodules via Section 4 ($A \cong B' \subseteq B$). Then*

$$\bar{F}_B(V) \;=\; \sum_{g \in T^S} V^S \;=\; T^S \cdot V^S, \qquad \bar{F}_A(V) \;=\; \sum_{g \in \mathrm{const}(T)} V^S \;=\; T \cdot V^S.$$

Both have the same *content power $(-)^S$ (direction set S); they differ* only *in the coproduct index, T^S for the weld against $T \cong \mathrm{const}(T)$ for the worker. Consequently $\bar{F}_A$ is a sub-prafunctor of $\bar{F}_B$, with $\bar{F}_A \cong \bar{F}_B$* iff *$|S| = 1$ or $|T| = 1$, and a* proper *inclusion exactly on $|S| \geq 2 \wedge |T| \geq 2$.*

Proof. $\bar{F}_B = \sum_{g \in T^S} V^S$ is Theorem 4: the coproduct index is the weld positions modulo the left $\mathsf{St}S$-collapse (the S-head is killed by indiscreteness), namely the branch maps T^S, and the surviving direction is S (the T-direction is killed by the right $\mathsf{St}T$-collapse). By Proposition 1, $A \cong B'$ is B restricted on positions to $\mathrm{const}(T) \subseteq T^S$ with directions unchanged; the same recipe restricts the coproduct index to $\mathrm{const}(T)$ and leaves the content power $(-)^S$ untouched, giving $\bar{F}_A = \sum_{t \in T} V^S = T \cdot V^S$.

Equality of the two functors for all V is equality of polynomials $T^S \mathsf{y}^S$ and $T\mathsf{y}^S$, i.e. $|T^S| = |T|$, i.e. $m^n = m$ where $n = |S|, m = |T|$. For $n, m \geq 1$ this holds iff $n = 1$ or $m = 1$. Otherwise $T \subsetneq T^S$ and the inclusion $\bar{F}_A \hookrightarrow \bar{F}_B$ is proper. □

Remark 3 (what this corrects). The exponents *agree* — both content powers are $(-)^S$. An earlier argument found "S versus $S \times T$", but that compared B under its right $\mathsf{St}T$-coaction against A read with *no* right coaction (the y-reading). With the coaction present, the T-directions of A collapse into the exponent S, not $S \times T$. The consistent separation is the coproduct index T versus T^S, nonzero precisely where non-constant branch maps exist, i.e. on $|S| \geq 2 \wedge |T| \geq 2$. In particular at $|S| = 1$ one has $T^S = T$, so $\mathrm{const}(T) = T^S$ and $\bar{F}_A = \bar{F}_B = T \cdot V$; moreover the carriers coincide, $B = \mathsf{St}S \triangleleft \mathsf{St}T = \mathsf{y} \triangleleft \mathsf{St}T = \mathsf{St}T = \mathsf{St}(1 \times T) = A$. There the worker *is* the content, on the nose, and no negative verdict may be claimed.

Verdict for clause (2). The identification "worker A = right-adjoint content factor of the weld" is **false on** $|S| \geq 2 \wedge |T| \geq 2$ — by a coproduct-index mismatch, not an exponent mismatch — and **true (indeed $A = B$) when** $|S| = 1$ **or** $|T| = 1$. The content power $(-)^S$ is *shared*; what the worker fails to be is the whole prafunctor.

Example 1. At $(|S|, |T|) = (2, 2)$, $V = 2$: $\bar{F}_A = 2 \cdot 2^2 = 8$ and $\bar{F}_B = (2 \cdot 2)^2 = 16$ (proper). At $(1, 2)$: $\bar{F}_A = \bar{F}_B = 4$; at $(1, 3)$: $\bar{F}_A = \bar{F}_B = 6$ (equal). These are exactly the shapes where the criterion "$|T| \geq 2$" and the corrected criterion "$|S| \geq 2 \wedge |T| \geq 2$" disagree.

6 The retract: what the worker forgets

The worker arises from the weld by an idempotent retract; we record honestly what it keeps and discards. The weld's coproduct index splits as

$$T^S \;=\; \mathrm{const}(T) \;\sqcup\; \mathrm{NC}, \qquad |\mathrm{const}(T)| = m, \quad |\mathrm{NC}| = m^n - m,$$

where NC is the set of non-constant maps and $n = |S|, m = |T|$. The retract is $e\colon S \times T^S \to S \times T^S$, $e(s,g) = (s, \mathrm{const}_{g(s)})$: collapse each branch map to the constant at its value on the current input.

Theorem 6 (Keep/discard)**.** *e is idempotent with image $S \times \mathrm{const}(T) \cong S \times T$, the worker positions $\sigma(A)$. For each current input s, e* keeps *the value $g(s) \in T$ (the diagonal T-coordinate) and* discards *the restriction $g|_{S\setminus\{s\}} \in T^{S\setminus\{s\}}$, a fibre of size m^{n-1}. Concretely the fibre of e over (s, const_t) is $\{g : g(s) = t\} \cong T^{S\setminus\{s\}}$.*

Proof. $e \circ e = e$ since $\mathrm{const}_{g(s)}$ evaluated at s is $g(s)$. The image is the constant slice $\{(s, \mathrm{const}_t)\}$. Two maps g, g' satisfy $e(s,g) = e(s,g')$ iff $g(s) = g'(s)$; fixing $g(s) = t$ leaves $S \setminus \{s\}$ free, so the fibre is $T^{S\setminus\{s\}}$, of size m^{n-1}. □

Re-decision of clause (3). As first stated — "the discarded data is precisely the non-constant branch maps, and that is the Σ-tail" — the claim is **false**, for three independent reasons.

(i) The Σ-tail index is *all* of $T^S = \mathrm{const}(T) \sqcup \mathrm{NC}$, not NC. "$T^S$ = the non-constant maps" is a cardinality contradiction ($m^n \neq m^n - m$).

(ii) What e forgets per position is the fibre $T^{S\setminus\{s\}}$ of size m^{n-1} — neither NC (size $m^n - m$) nor T^S (size m^n). At $(2,3)$: 3 versus 6 versus 9, all distinct.

(iii) The fibre $\{g : g(s) = t\}$ is not "the non-constant maps": it contains the one constant map const_t and omits every non-constant g with $g(s) \neq t$.

The **true residue** is this: the weld's Σ-tail *is* the full branch-map set $T^S = \mathrm{Hom}(S,T)$ — that half of clause (3) stands — and the retract e realises the projection of this index onto the worker's constant slice $\mathrm{const}(T) \cong T$ by "evaluate at the current input, then make constant". The genuinely discarded structure is, per input s, the off-diagonal dependence $g|_{S\setminus\{s\}}$. It is nonempty — the retract actually loses something — exactly on $|S| \geq 2 \wedge |T| \geq 2$, the same locus as the clause-(2) separation.

This also settles the old worry about a "reversal of adjointness" honestly: T^S is the index of the *left*-adjoint coproduct $\sum_{T^S} \dashv \Delta_{T^S}$, not right-adjoint content; the right-adjoint content $(-)^S$ is *kept* by both worker and weld; and the worker-versus-weld difference is purely how much of the coproduct index is retained (T versus T^S) — no adjoint-direction clash at all.

7 The unified statement

Theorem 7 (Worker = constant-branch slice of the weld)**.** *Let S, T be finite nonempty. With A read as the $(\mathsf{St}T, \mathsf{St}S)$-bicomodule $\sigma(A) = B' \subseteq B$ of constant branch maps ($\sigma^\sharp = \mathrm{id}_{S\times T}$),*

$$\bar{F}_A(V) = T \cdot V^S \;\subseteq\; \bar{F}_B(V) = T^S \cdot V^S$$

is the sub-coproduct over $\mathrm{const}(T) \subseteq T^S$*. Worker and weld* share *the right-adjoint content power $(-)^S$ and differ* only *in the coproduct index (T versus T^S). The retract $e(s,g) = (s, \mathrm{const}_{g(s)})$ projects T^S onto* $\mathrm{const}(T)$*, forgetting per input s the off-diagonal fibre $T^{S\setminus\{s\}}$. The three degeneracies — $\bar{F}_A = \bar{F}_B$, empty discard, $A = B$ — coincide and hold* iff $|S| = 1$ or $|T| = 1$*; the proper separation is exactly $|S| \geq 2 \wedge |T| \geq 2$.*

Orchestration reading. State-*entanglement* (the worker A) keeps a branch choice per output token $t \in T$; data-*aggregation* (the weld B) keeps a branch choice per response policy $g \in T^S$. Both read the whole input state — the shared content $(-)^S$ — but the weld aggregates over all policies and the worker only over constant ones. Keeping the two distinct is a semantic requirement, not a defect, and the exact locus where it matters is $|S|, |T| \geq 2$: with a single input or output there is nothing to entangle.

Grant relevance. The weld now has a precise home in Poly: it is a data-migration prafunctor $\mathsf{St}T\text{-}\mathsf{Set} \to \mathsf{St}S\text{-}\mathsf{Set}$, $X \mapsto (T \times X)^S$, with the branch policy aggregated by $\sum_{T^S}$. This places the *state* axis of the compositional-correctness program inside the same framed bicategory $\mathsf{Cat}^\sharp$ that houses the directed (retrofunctor) axis. But the worker is not that prafunctor's content: entanglement and aggregation are different collapses of one bicomodule, and the orchestration semantics must keep them apart.

8 Response to the referee

We record where each referee item is addressed.

- **M1** (clause 2 inconsistent comparison): Section 5. Both objects are now read as $(\mathsf{St}T, \mathsf{St}S)$-bicomodules; the exponents agree, the separation is the coproduct index T versus T^S, and the locus is $|S| \geq 2 \wedge |T| \geq 2$, with $A = B$ shown at $|S| = 1$.
- **M2** (clause 3 self-contradiction): Section 6. $T^S = \mathrm{const}(T) \sqcup \mathrm{NC}$ is stated; "discarded = non-constant maps" is refuted ($3 \neq 6 \neq 9$ at $(2, 3)$); the true per-position residue is $T^{S \setminus \{s\}}$ and the true Σ-tail is the full T^S.
- **M3** (σ-display): Proposition 1. $\sigma(A) = B'$ on $\{(s, \mathrm{const}_t)\}$, $\sigma^\sharp = \mathrm{id}_{S \times T}$, prafunctor $T \cdot V^S$.
- **S1** (provenance): the reference [1] is Spivak–Garner–Fairbanks, *Functorial Aggregation* (arXiv:2111.10968), *not* Niu–Spivak [2] (arXiv:2312.00990). Every SGF locator used here (Def. 4.2, Rmk. 4.3, Thm. 4.28, Props. 3.6/3.9/5.2/5.7, Ex. 5.6) was re-checked against the published text.
- **S2** (naming/handedness): $B = \mathsf{St}S \triangleleft \mathsf{St}T$ is renamed the *unit-composite* bicomodule (Lemma 2), not a regular bicomodule; the left/right coaction labels follow Definition 1, with λ the left $d = \mathsf{St}S$ coaction and ρ the right $c = \mathsf{St}T$ coaction, so $m \colon \mathsf{St}T \to \mathsf{St}S$.
- **S5** and nits: nonemptiness $S, T \neq \emptyset$ is a standing hypothesis; Δ is reserved for the diagonal, the costate comonoids written $\mathsf{St}S, \mathsf{St}T$.

Honest scope. Lemma 2 is standard; the contribution is Section 4 (the worker is the constant-branch sub-bicomodule, with carrier exactly A) together with the corrected clause-(2)/(3) analysis built on the peer-reviewed Theorem 4. The caveat of Remark 2 on the cobase-change description is the only open bookkeeping, and it is not used in any verdict. All cardinality witnesses quoted above were machine-checked.

References

[1] D. I. Spivak, R. Garner, and A. Fairbanks, *Functorial Aggregation*, arXiv:2111.10968, 2021.

[2] N. Niu and D. I. Spivak, *Polynomial Functors: A Mathematical Theory of Interaction*, arXiv:2312.00990, 2023.